Solar Activity & Space Weather Monitoring System

DS3294: DS Practice Team #16 — Mid-Project Progress Report — March 2026

Project Objectives

This project (Team #16) aims to design and implement a reproducible monitoring and analytics system for solar activity and space weather indicators, with three overarching goals: understanding the temporal variability of solar phenomena, characterising the coupling between solar drivers and geospace responses, and confronting the challenges of multi-source physical time-series data.

Core objectives from the project brief:

- Identify and ingest publicly available solar and space weather datasets (sunspot numbers, geomagnetic indices Kp/Dst, solar flare) from authoritative sources.
- Harmonise datasets with differing temporal resolutions, units, and reporting conventions.
- Build a modular cleaning pipeline handling missing observations, reporting gaps, and known artefacts.
- Aggregate and align indicators across hourly, daily, and monthly scales.
- Analyse relationships between solar indicators and space weather responses via time-series correlation.
- Build an interactive dashboard visualising individual indicators, cross-indicator lagged comparisons, and anomalous activity periods.

Intensification objectives (advanced requirements):

- Implement and compare smoothing or baseline estimation strategies and assess their impact on interpretation.
- Identify and characterise extreme solar/geomagnetic activity episodes in temporal context.
- Analyse lead-lag relationships between solar drivers and geomagnetic responses.
- Explicitly discuss data limitations: measurement uncertainty, index construction, observational coverage.
- Perform at least one iterative refinement of aggregation, indicator selection, or visual encodings based on exploratory findings.

Steps Taken

(1) Data Acquisition. `download_data.py` fetches: SILSO daily sunspot numbers (1818–present), SILSO monthly sunspot numbers (1749–present), GFZ Potsdam 3-hourly Kp index (1932–present), and NOAA SWPC real-time Kp (∼last 7 days). Large archives are downloaded once and cached; a `--refresh` flag updates only recent windows.

(2) Ingestion. `ingest.py` parses cached files into typed Pandas DataFrames with a `DatetimeIndex`, exposing clean accessor functions (`fetch_sunspots()`, `fetch_kp_index()`).

(3) Pipeline Processing. `pipeline.py` applies: linear interpolation for gaps ≤ 3 days; physical-range clipping of Kp to $[0, 9]$; 3-hourly Kp aggregation to daily mean/max/storm-hours; outer-join alignment of sunspot and Kp datasets; three smoothing strategies; and rolling z-score anomaly scoring.

(4) Analysis. `analysis.py` computes lagged cross-correlations (up to ±60 days), detects extreme events via sigma-thresholding, estimates solar cycle phase, and produces monthly statistics.

(5) Dashboard. `app.py` delivers a Plotly Dash app *Time Series*, *Correlation*, *Extreme Events*, and *Smoothing Comparison*, all linked by a shared date-range picker and parameter sliders.

(6) Iterative Refinement. Following initial exploration: daily resolution was chosen over monthly (monthly hides the 1–4 day CME transit lag); GFZ was preferred over NOAA API for historical Kp; and Savitzky-Golay was adopted after centred rolling means were found to introduce phase-shift artefacts near data boundaries.

Major Algorithms, Processes & Mathematics

Lagged Cross-Correlation. For each integer lag $\ell \in [-L, +L]$ (up to $L = 60$ days), Pearson r between sunspot series $\mathbf{x}$ and Kp series $\mathbf{y}$:

$$r(\ell) = \frac{\sum_t (x_t - \bar{x})(y_{t+\ell} - \bar{y})}{\sqrt{\sum_t (x_t - \bar{x})^2}\ \sqrt{\sum_t (y_{t+\ell} - \bar{y})^2}}$$

The peak lag $\ell^* = \arg\max |r(\ell)|$ reveals the CME propagation delay (expected 1–4 days).

Extreme Event Detection. Rolling 365-day mean μ_t and std σ_t yield a z-score:

$$z_t = \frac{s_t - \mu_t}{\sigma_t}, \quad \text{flagged if } z_t > \theta, \quad \theta \in [1.5,\ 4.0]$$

θ is adjustable via dashboard slider, satisfying the intensification requirement on extreme event characterisation.

Smoothing Strategy Comparison. Three baselines are computed and overlaid in the *Smoothing* tab to assess their interpretive impact (intensification requirement):

1. *27-day rolling mean* — solar rotation period baseline.
2. *365-day rolling mean* — annual baseline; introduces edge phase-shift.

Solar Cycle Phase. Fractional position in the ∼11-year cycle from minimum t_0:

$$\phi_t = \frac{(t - t_0) \bmod T}{T} \in [0, 1), \quad T = 11 \times 365.25 \text{ days}$$

Kp Daily Aggregation. Three-hourly readings reduced to: daily mean $\overline{Kp}$, daily maximum $Kp_{\max}$, and storm-hour count.

Current Status of Work

Completed:

- Multi-source data acquisition with incremental refresh logic.
- Full ingestion and cleaning layer with physical sanity checks.
- Complete `pipeline.py`: gap filling, resampling, alignment, three smoothing methods, rolling z-scores.
- Complete `analysis.py`: cross-correlation, extreme event detection, cycle phase, monthly stats.
- Four-tab Plotly Dash dashboard with global interactive controls.
- Version-controlled repository with `requirements.txt` and documented data-flow.
- Iterative refinement cycle (resolution and smoothing method selection) completed and documented.

Remaining:

- Formal data limitations section (Kp index construction, sunspot counting uncertainty, coverage gaps).
- Dst index ingestion for multi-index geomagnetic comparison.
- Unit tests and validation against published event catalogues.
- Final interpretation write-up and project report submission.

Team Work Distribution

Member	Responsibilities
Vanah Gupta	Data acquisition (`download_data.py`), source research, data limitations documentation
Shailendra Pratap Singh	Ingestion & pipeline (`ingest.py`, `pipeline.py`): cleaning, resampling, align time for different dataets, smoothing
Abhishek Menon	Analysis (`analysis.py`): lag correlation, anomaly detection, cycle phase, monthly stats
Mulumudi Dinesh Karthik	Dashboard (`app.py`): Plotly Dash layout, tab design, interactive controls

Expected Output

Per the DS3294 brief, the final deliverable is a **well-documented solar activity and space weather monitoring dashboard** demonstrating principled time-series data handling, multi-source integration, and informed interpretation of coupled physical systems. Specifically:

- A **fully reproducible Python system** runnable via `pip install` + two commands, covering the complete pipeline from raw data fetch to interactive dashboard.
- An **interactive dashboard** covering: individual solar/geomagnetic time series, lagged cross-indicator correlations, extreme event timelines, and smoothing strategy comparisons.
- A **smoothing comparison analysis** contrasting rolling-statistic vs. cycle-based (Savitzky-Golay) baselines and documenting their impact on identifying solar cycle peaks.
- A **lead–lag characterisation** of the sunspot–Kp coupling, quantifying the CME propagation delay and its statistical uncertainty.
- A **catalogued extreme-event log** of major geomagnetic storms and solar maxima, contextualised within the 11-year solar cycle, derived from automated z-score thresholding.
- An explicit **data limitations discussion** addressing Kp index construction, sunspot number measurement uncertainty, and multi-decadal observational coverage gaps.